

Same Scene, Different Story: Evaluating Robot Perception Across Scene Phases in the Wild

Anonymous

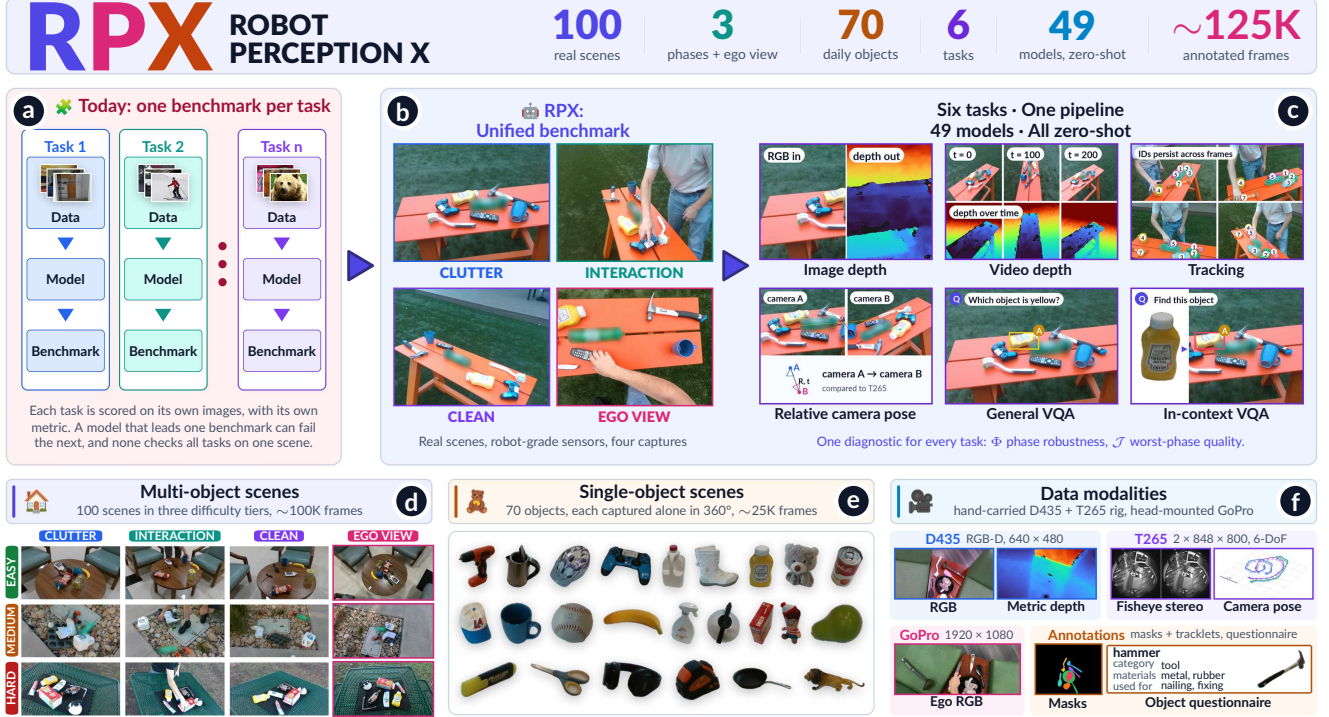


Fig. 1: We present **RPX**. (a) Today each perception task is benchmarked on its own data, usually internet imagery, with its own metric. (b) **RPX** evaluates six tasks on the same real scenes, captured on robot-grade sensors before (Clutter), during (Interaction) and after (Clean) manipulation, plus an egocentric view during Interaction. (c) One pipeline and one diagnostic for all tasks (Φ phase robustness, $\mathcal{J}$ worst-phase quality). (d) Multi-object scenes by difficulty (rows) and phase (columns), indoor and outdoor. (e) Each object is also captured alone in 360° [1]. (f) Sensors and annotation modalities.

Abstract—Robots change the scenes they perceive. A hand enters the frame, objects move, and the scene at the end differs from the start. Yet the perception models on these robots are chosen from benchmarks of frozen scenes. This paper asks how much that mismatch matters. We introduce **RPX**, a benchmark of 100 real scenes with 70 everyday objects, recorded with robot-grade sensors before, during, and after manipulation, 125K annotated frames. We evaluate 49 pretrained models zero-shot on six tasks: image depth, video depth, multi-object tracking, relative camera pose, and two forms of visual question answering. Two numbers summarize each model, phase robustness Φ from a repeated-measures MANOVA and worst-phase quality $\mathcal{J}_{\min}$. The results are not subtle. Every depth and tracking model and 20 of 24 VLM evaluations change significantly across phases ($p < 0.001$); near-range video-depth error grows 2–4 $\times$ while the robot acts, tracker identity switches grow 5–11 $\times$, and the best in-context grounding reaches only $\mathcal{J}_{\min} = 0.23$. Relative pose alone holds steady ($\Phi \geq 0.86$). Rankings also change from task to task. The model that tops a static leaderboard is not the one that keeps working once the robot moves; **RPX** tells them apart.

I. INTRODUCTION

Nearly every robot system built today rests on pretrained perception. Modular pipelines run depth estimators, trackers, and segmenters ahead of grasping and navigation [2]. Vision-language-action (VLA) policies inherit a frozen vision encoder, and the choice of that encoder is the largest single factor in how well the policy performs [3], [4], [5], [6]. Agentic systems call perception models again and again as they work through a task [7], [8], [9]. In each case the model was picked from a benchmark of still scenes, then put to work in scenes the robot itself rearranges.

That gap is easy to miss because it is spread across benchmarks. The depth model comes from the NYU Depth V2 leaderboard [10], the tracker from DAVIS [11], the vision-language model (VLM) from an internet question-answering leaderboard. Each was scored on a different camera and different scenes, and in all of them nothing moves. Some-

one assembling a manipulation stack has no way to check whether the models they chose survive a hand entering the frame, objects shifting, or occlusions that change mid-task. No existing benchmark records the same scene before, during, and after manipulation on robot-grade data (Table I).

We built **R_{PX}** (**R**obot **P**erception **X**; Fig. 1) to close that gap. Every scene is captured with robot-grade sensors in three phases, **Clutter** before manipulation, **Interaction** during it, and **Clean** after, with an egocentric stream during Interaction. Six tasks share the same physical scenes, and two numbers, phase robustness Φ and worst-phase quality $\mathcal{J}_{\min}$ (Sec. III-B), put every model on a common footing.

The main finding is that manipulation does not hurt all perception equally. Depth and tracking degrade sharply, relative pose barely moves, and the ranking of models changes from task to task and from one prompting mode to another on identical scenes. Section V works through this task by task; Section VI draws the cross-task lessons. Our contributions are:

- A **paired benchmark** of 100 real-world scenes captured before, during, and after manipulation on robot-grade sensors, with an egocentric Interaction stream.
- A **unified six-task evaluation** of 49 pretrained models (65 model–task evaluations), all zero-shot, scored by phase robustness Φ and worst-phase quality $\mathcal{J}_{\min}$.
- A **cross-task analysis** showing task-dependent fragility, ranking instability across phases and prompting modes, and robotics-critical failures that static benchmarks hide.

II. RELATED WORK

TABLE I: Comparison with benchmarks relevant to robot perception. ✓ = supported; ~ = partial; ✗ = not supported. ^{||}Policy-learning dataset. *Static extrinsics only. [†]Trajectories/trials. [‡]Across 7 core datasets.

	Perception					Robot Data		
	R_{PX}	ScanNet [12]	YCB-V [13]	Bop [14]	RoboDepth [15]	DROID [16]	Rt20T [17]	OK-Robot [2]
<i>Data properties</i>								
Real-world RGB-D	✓	✓	✓	✓	✗	✓	✓	✗
Metric depth GT	✓	✓	✓	✓	✗	✓	✓	✗
6-DoF camera pose	✓	✓	✓	✓	✗	✓	✓	✗
Calibrated stereo pairs	✓	✗	✓	✓	✗	✓	✗	✗
Instance masks	✓	✓	✓	✓	✗	✓	✗	✗
Object questionnaire	✓	✗	✗	✗	✗	✗	✗	✗
Multi-task GT on shared scenes	✓	✓	✗	✗	✗	✗	~	✗
<i>Protocol</i>								
Three-phase capture	✓	✗	✗	✗	✗	✓	✓	✗
Human interaction phase	✓	✗	✗	✗	✗	✓	✓	✗
Egocentric stream	✓	✗	✗	✗	✗	~	~	✗
Shared objects across phases	✓	✗	✗	✗	✗	✗	✗	✗
Single + multi-object scenes	✓	✗	✓	✓	✗	✗	✗	✗
Scene difficulty strata	✓	✗	✗	✗	✓	✗	✗	✗
# Scenes	100	1.5K	92	350 [§]	n/a	564	147	10
# Frames	125K	2.5M	134K	var.	n/a	76K [†]	110K [†]	171 [‡]
Focus	deploy.	recon.	pose	pose	depth	imit.	imit.	manip.

Pretrained models as robot perception. Pretrained vision models now do most of the seeing in robot systems. They serve as frozen encoders inside VLA policies [5], [4], [6], as preprocessing stages in modular pipelines [2], and as pseudo-label generators for world models [18]. Li et al. [3] found that which backbone a VLA uses matters more than any

other design choice, in simulation; on real hardware, adapting perception to the deployment scene through interactive teaching [19] directly improves manipulation. What is missing is a benchmark that evaluates these models together on real robot data. The **R_{PX}** benchmark closes this gap by evaluating them side by side on the same robot-captured scenes, before, during, and after manipulation.

Per-task benchmarks. Each perception task grew its own benchmark. Depth is scored on NYU Depth V2 [10], detection on COCO [20], odometry and depth on KITTI [21], each with its own camera, its own scenes, and its own domain. RoboDepth [15] tests depth robustness, but by corrupting existing benchmark images with synthetic blur, noise, and weather, so the perturbations are added after the fact rather than produced by anything happening in the scene. SenseShift6D [22] sweeps exposure and gain for 6-DoF pose, one task on static scenes. In **R_{PX}** the perturbation is the manipulation itself, recorded on the robot’s own sensors, and the same scenes serve all six tasks, so robustness numbers can be compared across tasks rather than read off six unrelated leaderboards.

Robot datasets. Table I lines **R_{PX}** up against the datasets a practitioner might reach for instead. YCB-Video [13] and BOP [14] give object pose ground truth in controlled setups where the scene never changes state. DROID [16] and RH20T [17] record manipulation at scale but carry no per-frame perception ground truth, so they cannot score a depth or tracking model. HOI4D [23] and HOT3D [24] record hand–object interaction from the wearer’s point of view with dense labels, but never capture the same scene before and after the interaction. GraspClutter6D [25] offers high-fidelity multi-sensor depth in heavy clutter, but the scenes are static. None of these follow one scene through manipulation with ground truth for several perception tasks, and none are built for the question **R_{PX}** asks. They exist to train policies or to score one capability. **R_{PX}** exists to choose a perception model for deployment, so its captures are paired, its tasks share scenes, and its scores share one diagnostic (Table I).

III. BENCHMARK DESIGN

A. Three-Phase Protocol

Every scene is recorded three times. In **Clutter** the scene is as we found it. In **Interaction** a person rearranges the objects by hand, as in learning-from-demonstration data [19], [16]. In **Clean** the scene has been reorganized. An exocentric RGB-D stream covers all three phases, and an egocentric stream covers Interaction (Sec. III-C).

B. Evaluation Primitives

A mean over metrics can hide a collapse on one behind stability on another, and it says nothing about whether the output changed when the scene did. We therefore reduce each model to two unit-free numbers, a phase-robustness effect size Φ and a worst-phase quality $\mathcal{J}_{\min}$.

Each task produces K metrics per (scene, phase) cell, forming a tensor $\mathbf{X} \in \mathbb{R}^{S \times P \times K}$ with $S=100$ scenes and

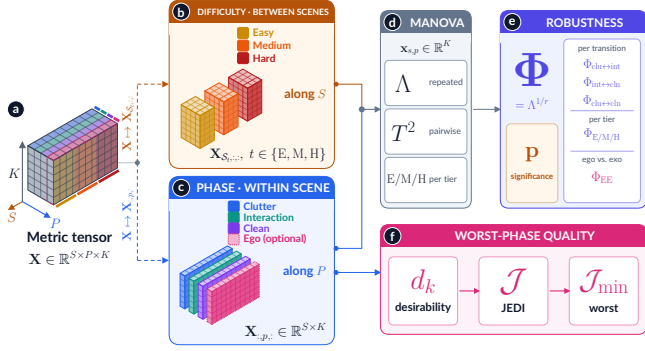


Fig. 2: Evaluation pipeline. (a) A model’s metric scores form a tensor over scenes, scene phases (ego dashed) and metrics. (b) Slices along S form the difficulty tiers; (c) slices along P are compared within scene and also feed the quality score. (d) A repeated-measures MANOVA gives Wilks’ Λ and per-transition tests, also run per tier. (e) Phase robustness Φ , overall, per transition and per difficulty tier. (f) Per-metric desirabilities combine into a per-phase quality $\mathcal{J}_p$; the worst phase, $\mathcal{J}_{\min}$, ranks models.

$P=3$ phases, plus the ego capture for T3 and T5–T6 (Fig. 2). Slicing $\mathbf{X}$ along P gives the per-phase $S \times K$ matrices that we compare within each scene; slicing along S gives the difficulty tiers $\mathcal{S}_t$, $t \in \{E, M, H\}$, on which every test below is repeated. For image depth $K=4$ (AbsRel, RMSE, SILog, δ_1 [26]); for tracking $K=3$ (HOTA [27], MOTA [28], IDSW); Table II lists K for every task. The metrics within a task are correlated but not redundant, so we test them jointly rather than one at a time.

State-Transition Robustness (Φ). We ask whether the K -metric vector changes across phases with a one-way repeated-measures MANOVA [29]. Let $\mathbf{x}_{s,p} \in \mathbb{R}^K$ be the z-scored, direction-aligned metric vector for scene s in phase p , and $\tilde{\mathbf{x}}_{s,p} = \mathbf{x}_{s,p} - \bar{\mathbf{x}}_{s,\cdot}$ the same vector centered on the scene’s mean over phases. From these we form two $K \times K$ sums-of-squares-and-cross-products (SSCP) matrices:

$$\mathbf{H} = S \sum_{p=1}^P \bar{\mathbf{x}}_{\cdot,p} \bar{\mathbf{x}}_{\cdot,p}^\top, \quad \mathbf{E} = \sum_{s,p} (\tilde{\mathbf{x}}_{s,p} - \bar{\mathbf{x}}_{s,\cdot})(\tilde{\mathbf{x}}_{s,p} - \bar{\mathbf{x}}_{s,\cdot})^\top \quad (1)$$

where $\bar{\mathbf{x}}_{\cdot,p}$ is the phase mean of the centered data. $\mathbf{H}$ holds the scatter between phases, the signal; $\mathbf{E}$ holds the scatter within phases, the noise. The effect size follows from Wilks’ Λ :

$$\Lambda = \frac{|\mathbf{E}|}{|\mathbf{H} + \mathbf{E}|}, \quad \eta_p^2 = 1 - \Lambda^{1/r}, \quad \Phi = 1 - \eta_p^2, \quad (2)$$

with $r = \min(K, P-1)$.

η_p^2 is the multivariate partial effect size, so Φ near 1 means the phases are indistinguishable. We take the univariate convention $\eta^2 \geq 0.14$ [30] as the reference line for a large effect, i.e. $\Phi \leq 0.86$. Significance uses Rao’s F -approximation of Λ [31] with $P-1$ hypothesis and $(S-1)(P-1)$ error degrees of freedom (DoF), assuming multivariate sphericity across phases; the resulting p tests the null of no phase effect. To see which transition is responsible (clu $\leftrightarrow$ int, for instance),

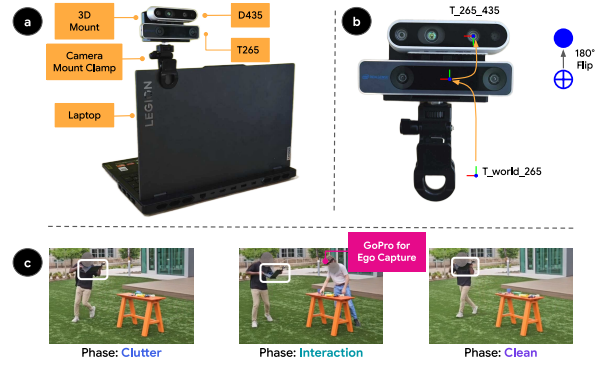


Fig. 3: Capture setup. (a) D435 + T265 rig on a laptop mount, carried by hand. (b) Sensor detail with the T265→D435 extrinsic and the world→T265 pose frames. (c) Three phases of one scene; the GoPro is visible during Interaction; faces blurred for anonymity.

we run a paired Hotelling T^2 [32] on the difference vectors $\mathbf{x}_{s,p_a} - \mathbf{x}_{s,p_b}$, Holm-corrected [33] over the $\binom{P}{2}$ pairs, which tells us whether a model breaks when the hand enters or only once the scene has been reorganized.

JEDI (Joint Empirical Desirability Index, $\mathcal{J}_{\min}$). Each metric k is mapped to a desirability $d_k = \text{clip}((v_k - v_k^{\text{worst}})/(v_k^{\text{best}} - v_k^{\text{worst}}), 0, 1)$, where v_k is the raw value and $(v_k^{\text{worst}}, v_k^{\text{best}})$ is a fixed window per metric, the same for every model (AbsRel 0.5→0, δ_1 0→1, HOTA 0→1, for example; the full list is in the toolkit). The quality of a phase is the geometric mean over its K metrics:

$$\mathcal{J}_p = \left(\prod_{k=1}^K d_{k,p} \right)^{1/K}, \quad \mathcal{J}_{\min} = \min_p \mathcal{J}_p \quad (3)$$

The geometric mean is deliberate. One collapsed metric drives $\mathcal{J}_p$ toward zero, as it would for a robot relying on it. $\mathcal{J}_{\min}$, the worst phase, is our primary ranking [34].

Φ and $\mathcal{J}$ are orthogonal by design. A model that is uniformly poor scores a trivially high Φ (≈ 1) and a low $\mathcal{J}$ (≈ 0). The model we want scores high on both. A high $\mathcal{J}_{\min}$ means it stays accurate in its worst phase, and a high Φ means that accuracy does not depend on the state of the scene.

C. Capture Protocol and Dataset

Manipulation has to be the only thing that changes between phases, so two people record every scene with one rig (Fig. 3). One carries the sensors, a RealSense D435 (RGB-D, 640×480) and a T265 (6-DoF visual-inertial odometry) on a shared laptop mount (Fig. 3a), and walks the scene by hand; the T265→D435 extrinsic and the world→T265 pose frames (Fig. 3b) give every D435 frame a metric camera pose. The other person rearranges the objects during Interaction and wears a head-mounted GoPro (1920×1080, RGB only, 6.8× the exocentric resolution) that records the egocentric stream. The rig is hand-held rather than on a tripod because a robot’s camera moves while it works. The same scene is then recorded before, during, and after the rearrangement (Fig. 3c) with the sensors and the object set unchanged, while

TABLE II: Phase-robustness summary. Top/bottom model per task ranked by $\mathcal{J}_{\min}$ (worst-phase quality). Φ : omnibus phase-robustness ($\Phi \leq 0.86$ = large effect). T4 uses 2 phases (clutter vs. clean); all other tasks use 3.

Task	K	M	Top model	$\mathcal{J}_{\min}\uparrow$	Bottom model	$\mathcal{J}_{\min}$	Φ (range)	Hardest transition	p
T1: Image Depth	4	10	Metric3D V2 [36]	0.71	Lotus-2 [37]	0.35	0.53–0.79	Clu→Int	<.001
T2: Video Depth	6	10	VGGT [38]	0.70	RollingDepth [39]	0.20	0.45–0.77	Clu→Int	<.001
T3: Tracking	3	11	SAM 2 Long [40]	0.96	SAM 3.1 [‡] [41]	0.72	0.50–0.75	Int (both)	<.001
T4: Rel. Pose	3	10	DA3 [42]	0.38	MonST3R [43]	0.08	0.86–0.97	Clu→Cln	>.05 (tier)
T5: Regular VQA	3	12	Qwen3-VL 8B [44]	0.57	Florence-2 B [45]	0.12	0.62–0.98	Clu→Cln	<.001
T6: In-Context VQA	3	12	Qwen3-VL 8B [44]	0.23	Florence-2 L [45]	0.01	0.65–0.96	Clu→Cln	<.001

viewpoint, framing, and lighting vary as they naturally do, so the phase factor tested in Sec. III-B is the manipulation and nothing else.

Difficulty tiers. We stratify scenes by *annotation-refinement effort*, the number of correction passes a scene’s masks needed, as a proxy for perceptual ambiguity. This gives Easy, Medium, and Hard splits of 33, 33, and 34 scenes.

Contents. 100 indoor and outdoor scenes (Fig. 1d), 70 objects, $\sim 125,000$ frames. The **multi-object stream** ($\sim 100K$ frames) is 100 scenes $\times$ 3 phases $\times$ ~ 250 D435 RGB-D frames, plus the egocentric GoPro during Interaction. Every frame carries RGB, metric depth, a 6-DoF pose (T265 VIO; validated in Sec. V), instance masks (SAM2 + GroundingDINO with human correction [35]), and object descriptions. The **single-object stream** ($\sim 24,500$ frames) records each object alone with RGB-D, pose, and calibrated stereo pairs.

IV. TASKS & MODELS

Tables III–VII list every model per task. Prompt variants (SAM 3.1 [41] box/text) count as separate evaluations; checkpoint variants (DA3 [42]) count once. Φ and $\mathcal{J}$ are computed on the metrics each field already uses on its dominant benchmark.

T1–T2 (Depth) are scored two ways, on a single RGB frame (*Image*) and on the full clip of a phase (*Video*), with the same ground truth and per-frame accuracy metrics; Video adds temporal gradient matching (TGM) and temporal gradient squared error (TGSE). **T3** (Tracking) gives each tracker the ground-truth masks of the first frame (8 semi-supervised trackers) or its boxes (SAM 3.1), or only an open-vocabulary text prompt ([‡]SAM 3.1, GSAM2). Interaction, where objects disappear behind a hand, is the stress test. **T4** (Relative Camera Pose Estimation, RCPE) scores $\sim 20K$ fixed image pairs per model with AUC@ τ , the standard pose-error AUC over the larger of the rotation error and the translation-direction error. **T5–T6** (VQA) test visual grounding with 6 VLM families at two sizes each (12 models). Every model answers with a bounding box through its native grounding tokens; a missing box is a complete miss (Acc 0, Center 0, GIoU at its floor of -1), so declining to answer earns nothing. The $K=3$ vector is Acc@0.5 (detection), GIoU [46] (spatial quality), and Center-in-GT (coarse localization).

T5 (Regular) asks the VLM to find the object a text description picks out, its color, material, or function from the per-object questionnaire (5 attribute question types, $\sim 12,000$ questions per model). T6 (In-Context) asks the same questions but replaces the description with a single reference

image of the object (the same 5 types, $\sim 6,000$ questions per model). T3 and T5–T6 are also run on the egocentric GoPro stream during Interaction, with ground-truth masks and tracklets for both streams, for a direct ego-versus-exo comparison at $6.8\times$ the resolution. Evaluating a new model requires only a thin adapter file; the `rpx-benchmark` toolkit computes Φ and $\mathcal{J}_{\min}$ directly from predictions, so **RPX** functions as a standing leaderboard, not a one-time study.

V. EXPERIMENTS

We evaluate every model zero-shot on 100 scenes $\times$ 3 phases (2 for relative camera pose, RCPE). Table II and Fig. 4 summarize the results; Φ and $\mathcal{J}$ are complete-case statistics with $n = 100$ scenes per cell.

A. T1: Image Depth

Every image depth model changes significantly across phases ($\Phi \leq 0.79$, all $p < 0.001$; Table III). Metric3D V2 has the best worst phase ($\mathcal{J}_{\min} = 0.71$, $\Phi = 0.68$). DA3 Metric-L is just as accurate (0.70) but the least stable model in the table ($\Phi = 0.53$, $\Phi_{\text{clu}\leftrightarrow\text{int}} = 0.23$), accurate and fragile at once. FE2E is the only relative-depth model whose $\mathcal{J}_{\min}$ (0.65) reaches the metric group. Even within one task, $\mathcal{J}_{\min}$ and Φ point to different models.

B. T2: Video Depth

VGGT has both the highest $\mathcal{J}_{\min}$ (0.70) and the highest Φ (0.77; Table IV). DA3 (video) matches it on overall quality ($\mathcal{J}_{\text{overall}} = 0.72$) but collapses during Interaction ($\Phi_{\text{clu}\leftrightarrow\text{int}} = 0.23$). The bottom three models, Video-DA, DepthCrafter, and RollingDepth, combine moderate Φ (0.54–0.58) with $\mathcal{J}_{\min} \leq 0.30$. For every video model, error at 0.3–1.0 m, the range at which a robot grasps, is $2\text{--}4\times$ higher during Interaction than in Clutter. Video depth is the one task where the best model is also the least fragile.

C. T3: Tracking

Every tracker degrades significantly ($\Phi \leq 0.75$, $p < 10^{-10}$; Table V), and the damage is specific to Interaction. Transitions into or out of it have $\Phi = 0.21\text{--}0.55$, while Clutter $\leftrightarrow$ Clean stays at $\Phi \geq 0.73$, so trackers recover once the hand leaves. SAM 2 Long has the highest $\mathcal{J}_{\min}$ (0.96). SAM 2 and SAM 2++ are the most stable of the mask-initialised trackers ($\Phi = 0.73$), and the open-vocabulary GSAM2 is the most stable overall ($\Phi = 0.75$) at much lower quality ($\mathcal{J}_{\min} = 0.79$). MiTS shows the largest phase effect ($\Phi = 0.63$) and the lowest $\mathcal{J}_{\min}$ (0.87); its identity switches

TABLE III: T1: Image depth ($K=4$: AbsRel, RMSE, SILog, δ_1). Out: D depth, P pose, N normals, GS Gaussian splats. Columns follow Fig. 2; bold Φ : best non- $\ddagger$ model. $\ddagger$ Relative depth, affine-aligned per phase, which can inflate Φ .

Model	Venue	Out	Params	Desirability $\mathcal{J} \uparrow$				Transition $\Phi \uparrow$			Scene-tier $\Phi \uparrow$			Summary	
				overall	clu	int	cln	clu $\leftrightarrow$ int	int $\leftrightarrow$ cln	clu $\leftrightarrow$ cln	E	M	H	$\mathcal{J}_{\min} \uparrow$	$\Phi \uparrow$
ZipDepth ‡ [47]	ECCV'26	D	6.14 M	0.56	0.58	0.51	0.60	0.42	0.37	0.54	0.63	0.59	0.46	0.51	0.60
HyDen ‡ [48]	ICLR'26	D	0.35 B	0.44	0.42	0.45	0.45	0.73	0.50	0.64	0.76	0.78	0.70	0.42	0.79
FE2E ‡ [49]	CVPR'26	D	12.47 B	0.68	0.70	0.65	0.68	0.39	0.41	0.49	0.62	0.55	0.49	0.65	0.59
DA3 Metric-L [42]	ICLR'26	D,P,N,GS	0.35 B	0.72	0.73	0.70	0.73	0.23	0.28	0.43	0.49	0.54	0.46	0.70	0.53
UniDepth V2 [50]	TPAMI'26	D	0.35 B	0.71	0.70	0.69	0.72	0.31	0.30	0.53	0.56	0.55	0.48	0.69	0.56
MoGe-2 ViT-L ‡ [51]	NeurIPS'25	D,P	0.33 B	0.52	0.48	0.54	0.53	0.43	0.51	0.51	0.66	0.66	0.61	0.48	0.66
Lotus-2 [37]	arXiv'25	D	17.58 B	0.36	0.35	0.38	0.35	0.34	0.66	0.44	0.64	0.61	0.55	0.35	0.64
Depth Pro [52]	ICLR'25	D	0.93 B	0.63	0.62	0.64	0.63	0.30	0.34	0.56	0.58	0.54	0.54	0.62	0.57
Metric3D V2 [36]	TPAMI'24	D,N	1.37 B	0.72	0.72	0.72	0.71	0.44	0.50	0.56	0.63	0.58	0.57	0.71	0.68
DA-V2 Large [53]	NeurIPS'24	D	0.34 B	0.60	0.60	0.55	0.66	0.27	0.33	0.43	0.59	0.49	0.51	0.55	0.54

TABLE IV: T2: Video depth ($K=6$: AbsRel, RMSE, SILog, δ_1 , TGM, TGSE). Same diagnostic format as Table III. All models evaluated on the full video of every capture (250 frames).

Model	Venue	Out	Params	Desirability $\mathcal{J} \uparrow$				Transition $\Phi \uparrow$			Scene-tier $\Phi \uparrow$			Summary	
				overall	clu	int	cln	clu $\leftrightarrow$ int	int $\leftrightarrow$ cln	clu $\leftrightarrow$ cln	E	M	H	$\mathcal{J}_{\min} \uparrow$	$\Phi \uparrow$
DA3 (video) [42]	ICLR'26	D,P,N,GS	0.41 B	0.72	0.78	0.64	0.74	0.23	0.26	0.46	0.46	0.42	0.37	0.64	0.48
Video-DA [54]	CVPR'25	D	0.38 B	0.33	0.35	0.35	0.30	0.21	0.55	0.32	0.57	0.49	0.44	0.30	0.54
ViGeo [55]	arXiv'26	D	1.25 B	0.64	0.63	0.63	0.67	0.23	0.42	0.35	0.53	0.54	0.48	0.63	0.54
VGGT [38]	CVPR'25	D,P,N	1.26 B	0.72	0.70	0.73	0.72	0.63	0.58	0.64	0.70	0.65	0.74	0.70	0.77
MonST3R [43]	ICLR'25	D,P	0.57 B	0.55	0.58	0.49	0.57	0.27	0.37	0.39	0.49	0.55	0.38	0.49	0.52
GemDepth [56]	ICML'26	D	0.71 B	0.69	0.76	0.61	0.71	0.17	0.31	0.34	0.46	0.39	0.37	0.61	0.45
DepthCrafter [57]	CVPR'25	D	2.25 B	0.30	0.31	0.31	0.26	0.33	0.54	0.30	0.60	0.49	0.47	0.26	0.57
DVD v1.1 [58]	arXiv'26	D	2.24 B	0.62	0.69	0.55	0.62	0.19	0.31	0.39	0.42	0.45	0.35	0.55	0.47
ChronoDepth [59]	CVPR'25	D	2.25 B	0.54	0.59	0.49	0.55	0.28	0.49	0.33	0.54	0.54	0.42	0.49	0.54
RollingDepth [39]	CVPR'25	D	1.29 B	0.22	0.23	0.24	0.20	0.27	0.59	0.34	0.60	0.54	0.47	0.20	0.58

rise 5–11 $\times$ during Interaction. SAM 3.1 with box prompts matches that $\mathcal{J}_{\min}$ with the lowest Φ of any tracker (0.50). Quality is uniformly high here, so Φ , not $\mathcal{J}_{\min}$, is what tells these trackers apart.

Ego-exo gap. The ego view leaves tracking quality almost where it was (SAM 2 Long and DAM4SAM hold at 0.96 and 0.95; SAM 2++ drops from $\mathcal{J}_{\text{exo,int}} = 0.92$ to $\mathcal{J}_{\text{ego}} = 0.86$), yet Φ_{EE} sits at or below the 0.86 line for 7 of 11 trackers, down to 0.55 for text-prompted SAM 3.1. With $\mathcal{J}$ nearly unchanged, the camera change alters the profile of the metric vector more than its level.

D. T4: Relative Camera Pose

We first check the ground truth. T265 poses agree with ARTag references on the 70 single-object captures to 1.53 $^\circ$ /4.68 cm (median), at least 4.5 $\times$ below the models' translation-direction errors (7–28 $^\circ$); rotation errors (0.7–1.3 $^\circ$) fall below that floor, so only translation direction is discriminative. RCPE is scored between Clutter and Clean only, because objects move during Interaction and pairwise relative pose assumes a static scene; two phases is a simplification, not a protocol limit. No model reaches significance within any tier ($p > 0.05$; $\Phi = 0.86$ –0.97; Table VI). DA3 leads ($\mathcal{J}_{\min} = 0.38$). DUST3R is the most phase-invariant ($\Phi = 0.97$) and second-worst in quality ($\mathcal{J}_{\min} = 0.16$). Relative pose is the one task where the state of the scene barely matters and quality alone ranks the models.

E. T5–T6: Visual Question Answering

Grounding is much harder than depth or tracking. The best $\mathcal{J}_{\min}$ is 0.57 on regular and 0.23 on in-context grounding (Table VII), below the best depth models (≈ 0.70) and every tracker (≥ 0.72). Qwen3-VL 8B leads both modes but

falls from $\mathcal{J} = 0.61$ to 0.27 in-context, and the order behind it does not carry over; DeepSeek-VL2 is ninth on regular and sixth on in-context, PaliGemma 2 3B seventh and ninth. The strongest VLMs are also the most phase-sensitive (Qwen3-VL 8B, $\Phi = 0.62$ regular, 0.65 in-context). Florence 2 reaches $\Phi = 0.94$ –0.98 only because it sits near the floor in every phase; it accounts for three of the four model-mode pairs that miss $\alpha = 0.001$, the fourth being PaliGemma 2 10B in-context. For VLMs a high Φ more often means a floor effect than robustness, so $\mathcal{J}_{\min}$ has to be read first.

Ego-exo gap. The head-mounted view helps grounding, so a low Φ_{EE} here marks the models that gain the most from it, Cosmos Reason2 8B ($\Phi_{\text{EE}} = 0.23$, $\mathcal{J}$ 0.52 $\rightarrow$ 0.67) and DeepSeek-VL2 4.5B (0.26, 0.41 $\rightarrow$ 0.53) on regular grounding, and again Cosmos Reason2 8B in-context (0.29, 0.11 $\rightarrow$ 0.33).

VI. DISCUSSION

We organize the discussion around three questions (RQ1–RQ3) and four findings that cut across tasks (F1–F4).

RQ1: How does perception vary with scene phase?

It depends on the task. Image depth ($\Phi = 0.53$ –0.79), video depth (0.45–0.77), and tracking (0.50–0.75) are phase-sensitive across the board. VQA spans a wider range, $\Phi = 0.62$ –0.98 for regular grounding and 0.65–0.96 for in-context grounding. Relative pose is largely phase-invariant ($\Phi = 0.86$ –0.97, Clutter–Clean only). Phase robustness is neither universal nor something one can read off task accuracy.

RQ2: Do rankings transfer across tasks and input conditions?

Only partly. Qwen3-VL 8B leads both VQA modes ($\mathcal{J}_{\min} = 0.57$ regular; 0.23 in-context), but the order behind it changes. Cosmos Reason2 8B drops from second (0.52) in

TABLE V: T3: Multi-object tracking ($K=3$: HOTA, MOTA, IDSW; IDF1 excluded due to $|r|>0.99$ collinearity with MOTA). GT first-frame masks; ego = GoPro, Interaction only. [‡]Text-prompted (no GT masks). Φ_{EE} : ego vs. exo-Interaction (camera, resolution and view confounded).

Model	Venue	Prompt	Params	Desirability $\mathcal{J} \uparrow$					Transition $\Phi \uparrow$				Scene-tier $\Phi \uparrow$			Summary	
				overall	clu	int	cln	ego	clu $\leftrightarrow$ int	int $\leftrightarrow$ cln	clu $\leftrightarrow$ cln	Φ_{EE}	E	M	H	$\mathcal{J}_{min} \uparrow$	$\Phi \uparrow$
SAM 2 Long [40]	ICCV'25	mask	224 M	0.98	0.98	0.96	0.99	0.96	0.42	0.40	0.94	0.78	0.71	0.66	0.60	0.96	0.69
DAM4SAM [60]	CVPR'25	mask	224 M	0.97	0.98	0.95	0.99	0.95	0.46	0.45	0.93	0.86	0.73	0.68	0.62	0.95	0.71
EdgeTAM [61]	CVPR'25	mask	14 M	0.96	0.97	0.93	0.98	0.92	0.43	0.45	0.94	0.96	0.67	0.74	0.65	0.93	0.71
SAM 2++ [62]	arXiv'25	mask	101 M	0.96	0.98	0.92	0.98	0.86	0.43	0.48	0.98	0.66	0.70	0.76	0.66	0.92	0.73
SAM 2 [35]	ICLR'25	mask	224 M	0.97	0.98	0.95	0.99	0.93	0.46	0.48	0.96	0.91	0.74	0.79	0.61	0.95	0.73
Cutie [63]	CVPR'24	mask	35 M	0.96	0.97	0.93	0.98	0.94	0.39	0.36	0.91	0.88	0.68	0.60	0.66	0.93	0.67
SAM 3.1 [41]	2026	bbox	873 M	0.94	0.97	0.87	0.97	0.90	0.21	0.21	0.93	0.71	0.53	0.49	0.41	0.87	0.50
MiTS [64]	ICCV'23	mask	33 M	0.93	0.95	0.87	0.97	0.86	0.44	0.31	0.73	0.80	0.62	0.67	0.54	0.87	0.63
XMem [65]	ECCV'22	mask	62 M	0.94	0.96	0.89	0.98	0.91	0.42	0.34	0.88	0.92	0.65	0.62	0.65	0.89	0.66
SAM 3.1 [‡] [41]	2026	text	873 M	0.79	0.83	0.72	0.82	0.74	0.29	0.35	0.96	0.55	0.66	0.58	0.52	0.72	0.62
GSAM2 [‡] [66], [67]	arXiv'24	text	397 M	0.85	0.87	0.79	0.90	0.86	0.55	0.50	0.93	0.75	0.84	0.72	0.67	0.79	0.75

TABLE VI: T4: RCPE ($K=3$: AUC@5°, @10°, @20°). Exo only (D435 + T265 pose GT; GoPro lacks pose GT). Φ and $\mathcal{J}$ compare clutter vs. clean (2 phases; single transition clu $\leftrightarrow$ cln).

Model	Venue	Out	Params	Desirability $\mathcal{J} \uparrow$			Transition clu $\leftrightarrow$ cln	Scene-tier $\Phi \uparrow$			Summary	
				overall	clu	cln		E	M	H	$\mathcal{J}_{\min} \uparrow$	$\Phi \uparrow$
DA3 [42]	ICLR'26	D,P,N,GS	1.36 B	0.42	0.38	0.45	0.87	0.87	0.89	0.86	0.38	0.87
VGGT [38]	CVPR'25	D,P,N	1.26 B	0.37	0.34	0.41	0.91	0.96	0.94	0.85	0.34	0.91
Reloc3R [68]	CVPR'25	P	0.43 B	0.40	0.37	0.42	0.91	0.92	0.89	0.91	0.37	0.91
Pi3X [69]	arXiv'25	P	0.99 B	0.38	0.34	0.41	0.93	0.93	0.93	0.92	0.34	0.93
MonST3R [43]	ICLR'25	D,P	0.27 B	0.09	0.08	0.11	0.86	0.90	0.81	0.86	0.08	0.86
MUST3R [70]	CVPR'25	P	0.42 B	0.27	0.24	0.29	0.92	0.91	0.93	0.92	0.24	0.92
Fast3R [71]	CVPR'25	P	0.65 B	0.28	0.26	0.30	0.95	0.95	0.95	0.95	0.26	0.95
CUT3R [72]	CVPR'25	D,P	0.46 B	0.31	0.28	0.34	0.91	0.95	0.86	0.92	0.28	0.91
MAST3R [73]	ECCV'24	P	0.69 B	0.34	0.32	0.37	0.96	0.97	0.97	0.93	0.32	0.96
DUST3R [74]	CVPR'24	P	0.57 B	0.18	0.16	0.19	0.97	0.95	0.99	0.96	0.16	0.97

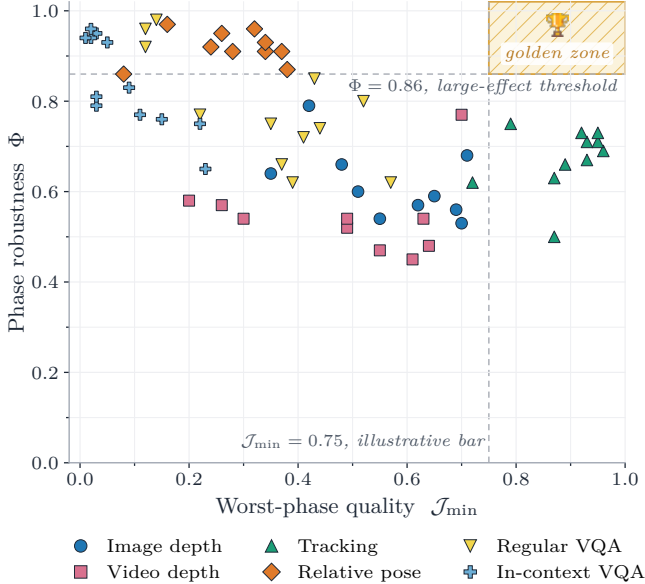


Fig. 4: Φ vs. $\mathcal{J}_{min}$ for all 65 model $\times$ task evaluations. Pose (♦) clusters at or above $\Phi = 0.86$ (small Clutter–Clean effects); tracking (▲) achieves high quality but degrades; depth and VQA spread across the fragile region. Dashed: $\Phi = 0.86$ and an illustrative $\mathcal{J}_{min} = 0.75$ bar; none of the 65 evaluations falls in the upper-right region that would mark a model as both accurate and robust within its task.

regular grounding to fourth (0.11) in in-context grounding, while Qwen3-VL 2B climbs from fourth (0.43) to second (0.22). Across tasks, VGGT leads video depth ($\mathcal{J}_{min} = 0.70$) but is only joint third for relative pose (0.34). A model has

to be chosen for the task and the conditioning mode it will actually face.

RQ3: How does the egocentric view affect performance?

It depends on the model. In regular grounding, ego $\mathcal{J}$ exceeds exocentric-Interaction $\mathcal{J}$ for all 12 VLMs. In in-context grounding, eight improve, one is unchanged, and three decline at the reported precision. Φ_{EE} ranges from 0.23–0.71 for regular grounding and 0.29–0.94 for in-context grounding; tracking spans $\Phi_{EE} = 0.55$ –0.96. Camera, resolution, and viewpoint change together in this comparison, so we treat it as descriptive rather than causal.

F1: Interaction is the principal stress condition for depth and tracking. Clutter–Interaction is the hardest transition for 15 of the 20 depth models. Tracking shows the same pattern; transitions involving Interaction have $\Phi = 0.21$ –0.55, whereas Clutter–Clean reaches 0.73–0.98. The large phase shifts come from active manipulation, not from recapturing a static scene.

F2: In-context VQA exposes a localization bottleneck, not just an output-generation failure. Seven of the 12 VLMs return a valid box on at least 90% of the in-context attribute questions, yet the best Acc@0.5 is only 0.215 and the best $\mathcal{J}_{min}$ is 0.23. Regular grounding reaches 0.593 and 0.57. A valid box is not a correct one. In-context grounding asks the model to read attributes off a reference image and bind them to the right instance in the scene, which is harder than localizing a named object, especially when several candidates share a color, material, or function.

F3: Robustness and quality are complementary. A high Φ can mean uniformly poor predictions rather than useful robustness. Qwen3-VL 8B pairs high regular-grounding quality

TABLE VII: T5–T6: VQA localization ($K=3$: Acc@0.5, GlOUI, Center-in-GT). Regular: 5 attribute question types, $\sim 12,000$ questions per model; in-context: the same questions with the subject given as a reference image instead of text, $\sim 6,000$ questions. Each block has its own per-scene MANOVA. † MoE active parameters. Φ_{EE} as in Table V.

Model	Params	Desirability $\mathcal{J} \uparrow$					Transition $\Phi \uparrow$				Scene-tier $\Phi \uparrow$			Summary	
		overall	clu	int	cln	ego	clu $\leftrightarrow$ int	int $\leftrightarrow$ cln	clu $\leftrightarrow$ cln	Φ_{EE}	E	M	H	$\mathcal{J}_{min} \uparrow$	$\Phi \uparrow$
T5: Regular grounding															
Cosmos Reason2 [75]	8 B	0.54	0.55	0.52	0.54	0.67	0.75	0.63	0.58	0.23	0.70	0.80	0.80	0.52	0.80
Cosmos Reason2 [75]	2 B	0.25	0.28	0.26	0.22	0.38	0.64	0.86	0.45	0.48	0.76	0.72	0.77	0.22	0.77
Qwen3-VL [44]	8 B	0.61	0.63	0.57	0.61	0.69	0.31	0.80	0.35	0.48	0.55	0.56	0.67	0.57	0.62
Qwen3-VL [44]	2 B	0.46	0.43	0.49	0.45	0.62	0.85	0.71	0.62	0.50	0.74	0.85	0.84	0.43	0.85
InternVL 3.5 [76]	14 B	0.48	0.52	0.44	0.49	0.55	0.59	0.74	0.45	0.53	0.70	0.70	0.76	0.44	0.74
InternVL 3.5 [76]	1 B	0.42	0.45	0.39	0.43	0.41	0.43	0.63	0.32	0.71	0.60	0.60	0.64	0.39	0.62
PaliGemma 2 [77]	10 B	0.38	0.43	0.35	0.37	0.42	0.49	0.87	0.50	0.60	0.69	0.77	0.76	0.35	0.75
PaliGemma 2 [77]	3 B	0.39	0.42	0.37	0.39	0.44	0.50	0.66	0.33	0.67	0.66	0.64	0.66	0.37	0.66
DeepSeek-VL2 [78]	4.5 B [†]	0.48	0.54	0.41	0.48	0.53	0.47	0.64	0.56	0.26	0.72	0.66	0.70	0.41	0.72
DeepSeek-VL2 [78]	1.0 B [†]	0.15	0.19	0.12	0.14	0.19	0.77	0.98	0.82	0.44	0.90	0.94	0.87	0.12	0.92
Florence 2 [45]	0.77 B	0.15	0.16	0.15	0.14	0.16	0.99	0.95	0.95	0.65	0.96	0.94	0.98	0.14	0.98
Florence 2 [45]	0.23 B	0.13	0.14	0.12	0.12	0.16	0.91	0.99	0.90	0.69	0.98	0.92	0.92	0.12	0.96
T6: In-context grounding															
Cosmos Reason2 [75]	8 B	0.13	0.15	0.11	0.14	0.33	0.50	0.96	0.47	0.29	0.79	0.73	0.73	0.11	0.77
Cosmos Reason2 [75]	2 B	0.10	0.12	0.09	0.09	0.21	0.66	0.97	0.51	0.58	0.72	0.86	0.83	0.09	0.83
Qwen3-VL [44]	8 B	0.27	0.31	0.23	0.28	0.34	0.40	0.74	0.35	0.63	0.57	0.70	0.64	0.23	0.65
Qwen3-VL [44]	2 B	0.23	0.25	0.22	0.22	0.35	0.67	0.76	0.37	0.67	0.73	0.74	0.73	0.22	0.75
InternVL 3.5 [76]	14 B	0.19	0.24	0.15	0.17	0.12	0.58	0.86	0.46	0.67	0.68	0.75	0.79	0.15	0.76
InternVL 3.5 [76]	1 B	0.05	0.07	0.03	0.04	0.02	0.64	0.82	0.52	0.61	0.76	0.83	0.72	0.03	0.79
PaliGemma 2 [77]	10 B	0.03	0.04	0.03	0.03	0.09	0.95	0.93	0.85	0.49	0.95	0.86	0.96	0.03	0.95
PaliGemma 2 [77]	3 B	0.04	0.04	0.03	0.05	0.07	0.53	0.83	0.73	0.73	0.76	0.79	0.84	0.03	0.81
DeepSeek-VL2 [78]	4.5 B [†]	0.07	0.10	0.05	0.05	0.10	0.83	1.00	0.83	0.75	0.89	0.96	0.88	0.05	0.93
DeepSeek-VL2 [78]	1.0 B [†]	0.02	0.03	0.02	0.02	0.01	0.84	0.99	0.84	0.94	0.92	0.95	0.87	0.02	0.94
Florence 2 [45]	0.77 B	0.01	0.02	0.01	0.01	0.02	0.81	0.95	0.91	0.88	0.90	0.91	0.90	0.01	0.94
Florence 2 [45]	0.23 B	0.02	0.02	0.02	0.02	0.02	0.87	0.98	0.92	0.72	0.93	0.89	0.93	0.02	0.96

($\mathcal{J}_{min} = 0.57$) with relatively low robustness ($\Phi = 0.62$), while Florence 2 reaches $\Phi = 0.94$ – 0.98 without leaving the quality floor ($\mathcal{J}_{min} = 0.01$ – 0.14). Rank by $\mathcal{J}_{min}$ first, then screen for fragility with Φ .

F4: Prompt modality changes both tracking quality and robustness. SAM 3.1 initialized with boxes reaches $\mathcal{J}_{min} = 0.87$ at $\Phi = 0.50$; with text prompts, quality drops to $\mathcal{J}_{min} = 0.72$ and robustness rises to $\Phi = 0.62$. The prompt is part of the configuration under test, and the higher-quality variant is not necessarily the more phase-invariant one.

VII. LIMITATIONS

RPX has three limits of scope. First, the exocentric rig is carried by a person, and the ego–exo comparison changes viewpoint, camera, and resolution at once, so it cannot isolate embodiment or viewpoint effects. Second, the depth reference comes from the D435’s active stereo, which hands and occlusions during Interaction may themselves disturb. The paired design supports comparisons between models, but absolute depth changes in the Interaction phase should be read with care. Relative pose is likewise limited to Clutter–Clean because Interaction breaks its static-scene assumption. Third, RPX measures zero-shot perception quality and phase robustness, not end-to-end robot success.

VIII. CONCLUSION

RPX measures what happens to a perception model when the robot changes the scene in front of it. It records the same scene before, during, and after manipulation and scores 49 models on six tasks with two numbers, Φ and $\mathcal{J}_{min}$. Every depth and tracking model and 20 of 24 VLM evaluations shift significantly across phases ($p < 0.001$), while relative pose barely moves ($\Phi \geq 0.86$). Rankings change from task

to task, between VQA modes, and with the prompt, on identical scenes. The failures that matter most to a robot, video-depth error rising 2–4 $\times$ at 0.3–1.0 m, identity switches growing 5–11 $\times$, and in-context grounding topping out at $\mathcal{J}_{min} = 0.23$, do not show up on a static benchmark at all. A leaderboard of frozen scenes leaves the hardest operating regime untested; **RPX** tests it. Future work will add new tasks on the modalities we already capture, stereo depth from the calibrated pairs and novel view synthesis from the T265-posed frames, open the toolkit so the community can try their own models, and relate $\mathcal{J}_{min}$ to manipulation success. Data, code, and the `rpx-benchmark` toolkit will be released at <https://anonymous.4open.science/r/rpx-anonymous>; the data is withheld until acceptance to preserve anonymity.

REFERENCES

- [1] J. J. P. *et al.*, “FewSQL: A Dataset for Few-Shot Object Learning in Robotic Environments,” in *ICRA*, 2023.
- [2] P. Liu *et al.*, “OK-Robot: What really matters in integrating open-knowledge models for robotics,” in *RSS*, 2024.
- [3] X. Li *et al.*, “What matters in building vision–language–action models for generalist robots,” *Nature Machine Intelligence*, vol. 8, no. 2, pp. 158–172, 2026.
- [4] K. Black *et al.*, “ π_0 : A vision-language-action flow model for general robot control,” in *RSS*, 2025.
- [5] M. J. Kim *et al.*, “OpenVLA: An open-source vision-language-action model,” in *CoRL*, 2024.
- [6] J. Bjorck *et al.*, “GR00T N1: An open foundation model for generalist humanoid robots,” *arXiv preprint arXiv:2503.14734*, 2025.
- [7] W. Huang *et al.*, “VoxPoser: Composable 3D value maps for robotic manipulation with language models,” in *CoRL*, 2023.
- [8] M. Ahn *et al.*, “Do as i can, not as i say: Grounding language in robotic affordances,” in *CoRL*, 2022.
- [9] R. Lu *et al.*, “ASPIRE: Agentic skills discovery for robotics,” *arXiv preprint arXiv:2607.00272*, 2026.
- [10] N. Silberman *et al.*, “Indoor segmentation and support inference from RGBD images,” in *ECCV*, 2012.

- [11] F. Perazzi *et al.*, “A benchmark dataset and evaluation methodology for video object segmentation,” in *CVPR*, 2016.
- [12] A. Dai *et al.*, “ScanNet: Richly-annotated 3D reconstructions of indoor scenes,” in *CVPR*, 2017.
- [13] Y. Xiang *et al.*, “PoseCNN: A convolutional neural network for 6D object pose estimation in cluttered scenes,” in *RSS*, 2018.
- [14] T. Hodaň *et al.*, “BOP challenge 2023 on detection, segmentation and pose estimation of seen and unseen rigid objects,” *arXiv preprint arXiv:2403.09799*, 2024.
- [15] L. Kong *et al.*, “RoboDepth: Robust out-of-distribution depth estimation under corruptions,” in *NeurIPS*, vol. 36, 2023.
- [16] A. Khazatsky *et al.*, “DROID: A large-scale in-the-wild robot manipulation dataset,” in *RSS*, 2024.
- [17] H.-S. Fang *et al.*, “RH20T: A comprehensive robotic dataset for learning diverse skills in one-shot,” in *ICRA*, 2024.
- [18] H. Zhen *et al.*, “Tesseract: Learning 4D embodied world models,” in *ICCV*, 2025.
- [19] J. J. P. *et al.*, “iTeach: In the wild interactive teaching for failure-driven adaptation of robot perception,” *arXiv:2410.09072*, 2026.
- [20] T.-Y. Lin *et al.*, “Microsoft COCO: Common objects in context,” in *ECCV*, 2014.
- [21] A. Geiger *et al.*, “Are we ready for autonomous driving? the KITTI vision benchmark suite,” in *CVPR*, 2012.
- [22] Y. Han *et al.*, “SenseShift6D: Multimodal RGB-D benchmarking for robust 6D pose estimation across environment and sensor variations,” *arXiv preprint arXiv:2507.05751*, 2025.
- [23] Y. Liu *et al.*, “HOI4D: A 4D egocentric dataset for category-level human-object interaction,” in *CVPR*, 2022.
- [24] P. Banerjee *et al.*, “HOT3D: Hand and object tracking in 3D from egocentric multi-view videos,” in *CVPR*, 2025.
- [25] S. Back *et al.*, “GraspClutter6D: A large-scale real-world dataset for robust perception and grasping in cluttered scenes,” *IEEE Robotics and Automation Letters*, 2025.
- [26] D. Eigen *et al.*, “Depth map prediction from a single image using a multi-scale deep network,” in *NeurIPS*, 2014.
- [27] J. Luiten *et al.*, “HOTA: A higher order metric for evaluating multi-object tracking,” *IJCV*, vol. 129, pp. 548–578, 2021.
- [28] K. Bernardin *et al.*, “Evaluating multiple object tracking performance: The CLEAR MOT metrics,” *EURASIP Journal on Image and Video Processing*, vol. 2008, pp. 1–10, 2008.
- [29] S. S. Wilks, “Certain generalizations in the analysis of variance,” *Biometrika*, vol. 24, no. 3/4, pp. 471–494, 1932.
- [30] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Lawrence Erlbaum, 1988.
- [31] C. R. Rao, “An asymptotic expansion of the distribution of Wilks’ criterion,” *Bulletin of the International Statistical Institute*, vol. 33, no. 2, pp. 177–180, 1951.
- [32] H. Hotelling, “The generalization of Student’s ratio,” *Annals of Mathematical Statistics*, vol. 2, no. 3, pp. 360–378, 1931.
- [33] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [34] E. C. Harrington, “The desirability function,” *Industrial Quality Control*, vol. 21, no. 10, pp. 494–498, 1965.
- [35] N. Ravi *et al.*, “SAM 2: Segment anything in images and videos,” in *ICLR*, 2025.
- [36] M. Hu *et al.*, “Metric3D v2: A versatile monocular geometric foundation model for zero-shot metric depth and surface normal estimation,” *IEEE TPAMI*, vol. 46, no. 12, pp. 10 579–10 596, 2024.
- [37] J. He *et al.*, “Lotus-2: Advancing geometric dense prediction with powerful image generative model,” *arXiv preprint arXiv:2512.01030*, 2025.
- [38] J. Wang *et al.*, “VGGT: Visual geometry grounded transformer,” in *CVPR*, 2025.
- [39] B. Ke *et al.*, “Video depth without video models,” in *CVPR*, 2025.
- [40] S. Ding *et al.*, “SAM2Long: Enhancing SAM 2 for long video segmentation with a training-free memory tree,” in *ICCV*, 2025.
- [41] Meta AI, “SAM 3.1: Faster and more accessible real-time video detection and tracking with multiplexing and global reasoning,” Meta AI model update, Mar. 2026.
- [42] H. Lin *et al.*, “Depth anything 3: Recovering the visual space from any views,” in *ICLR*, 2026.
- [43] J. Zhang *et al.*, “MonST3R: A simple approach for estimating geometry in the presence of motion,” in *ICLR*, 2025.
- [44] S. Bai *et al.*, “Qwen3-VL technical report,” *arXiv preprint arXiv:2511.21631*, 2025.
- [45] B. Xiao *et al.*, “Florence-2: Advancing a unified representation for a variety of vision tasks,” in *CVPR*, 2024.
- [46] H. Rezatofighi *et al.*, “Generalized intersection over union: A metric and a loss for bounding box regression,” in *CVPR*, 2019.
- [47] F. Tosi *et al.*, “ZipDepth: Bringing lightweight zero-shot monocular depth anywhere, on any device,” in *ECCV*, 2026.
- [48] Z. Zhang *et al.*, “Hyden: A hybrid dual-path encoder for monocular geometry of high-resolution images,” in *ICLR*, 2026.
- [49] J. Wang *et al.*, “FE2E: From editor to dense geometry estimator,” in *CVPR*, 2026.
- [50] L. Piccinelli *et al.*, “UniDepthV2: Universal monocular metric depth estimation made simpler,” *IEEE TPAMI*, vol. 48, no. 3, pp. 2354–2367, 2026.
- [51] R. Wang *et al.*, “MoGe-2: Accurate monocular geometry with metric scale and sharp details,” in *NeurIPS*, vol. 38, 2025.
- [52] A. Bochkovskii *et al.*, “Depth pro: Sharp monocular metric depth in less than a second,” in *ICLR*, 2025.
- [53] L. Yang *et al.*, “Depth anything V2,” in *NeurIPS*, 2024.
- [54] S. Chen *et al.*, “Video depth anything: Consistent depth estimation for super-long videos,” in *CVPR*, 2025.
- [55] Z. Yu *et al.*, “Towards consistent video geometry estimation,” *arXiv preprint arXiv:2605.30060*, 2026.
- [56] Y. Liu *et al.*, “Gemdepth: Geometry-embedded features for 3D-consistent video depth,” in *ICML*, 2026.
- [57] W. Hu *et al.*, “Depthrafter: Generating consistent long depth sequences for open-world videos,” in *CVPR*, 2025.
- [58] H. Zhang *et al.*, “DVD: Deterministic video depth estimation with generative priors,” *arXiv preprint arXiv:2603.12250*, 2026.
- [59] J. Shao *et al.*, “Learning temporally consistent video depth from video diffusion priors,” in *CVPR*, 2025.
- [60] J. Videnovic *et al.*, “A distractor-aware memory for visual object tracking with SAM2,” in *CVPR*, 2025.
- [61] C. Zhou *et al.*, “EdgeTAM: On-device track anything model,” in *CVPR*, 2025.
- [62] J. Zhang *et al.*, “SAM 2++: Tracking anything at any granularity,” *arXiv preprint arXiv:2510.18822*, 2025.
- [63] H. K. Cheng *et al.*, “Putting the object back into video object segmentation,” in *CVPR*, 2024.
- [64] Y. Xu *et al.*, “Integrating boxes and masks: A multi-object framework for unified visual tracking and segmentation,” in *ICCV*, 2023.
- [65] H. K. Cheng *et al.*, “XMem: Long-term video object segmentation with an Atkinson-Shiffrin memory model,” in *ECCV*, 2022.
- [66] S. Liu *et al.*, “Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection,” in *ECCV*, 2024.
- [67] T. Ren *et al.*, “Grounded SAM 2: Ground and track anything in videos,” Official IDEA-Research software release, 2024.
- [68] S. Dong *et al.*, “Reloc3r: Large-scale training of relative camera pose regression for generalizable, fast, and accurate visual localization,” in *CVPR*, 2025.
- [69] Y. Wang *et al.*, “Pi3X model release,” Official π^3 project repository, Dec. 2025.
- [70] Y. Cabon *et al.*, “MUST3R: Multi-view network for stereo 3D reconstruction,” in *CVPR*, 2025.
- [71] J. Yang *et al.*, “Fast3R: Towards 3D reconstruction of 1000+ images in one forward pass,” in *CVPR*, 2025.
- [72] Q. Wang *et al.*, “Continuous 3D perception model with persistent state,” in *CVPR*, 2025.
- [73] V. Leroy *et al.*, “Grounding image matching in 3D with MAST3R,” in *ECCV*, 2024.
- [74] S. Wang *et al.*, “DUST3R: Geometric 3D vision made easy,” in *CVPR*, 2024.
- [75] NVIDIA, “Cosmos-Reason2: Physical AI common sense and embodied reasoning models,” <https://huggingface.co/nvidia/Cosmos-Reason2-8B>, 2025.
- [76] W. Wang *et al.*, “InternVL3.5: Advancing open-source multimodal models in versatility, reasoning, and efficiency,” *arXiv preprint arXiv:2508.18265*, 2025.
- [77] A. Steiner *et al.*, “PaliGemma 2: A family of versatile VLMs for transfer,” *arXiv preprint arXiv:2412.03555*, 2024.
- [78] Z. Wu *et al.*, “DeepSeek-VL2: Mixture-of-experts vision-language models for advanced multimodal understanding,” *arXiv preprint arXiv:2412.10302*, 2024.